BIO 004 · HUMAN ANATOMY · SOLANO COLLEGE

Module 3 Lab Packet.

Muscle · Head, Neck, Trunk, and Upper Limb Muscles · Heart · Vessels · Nerves · Blood

PART A	**Lab structure list** Every structure you can be asked to identify on the Module 3 lab exam, chapter by chapter.
PART B	**Muscle charts** Origin, insertion, action, and nerve for every muscle in this unit, including the face and neck.
PART C	**Blood cell charts** What each cell looks like, its cell line, its job, when it runs high or low, and the terms for each.
PART D	**Lab exam questions** The kinds of questions you could be asked at a station, by chapter.

Dr. Sharilyn Rennie

MODULE 3 · LAB

How to use this packet

In lab

Bring Part A to every lab. Check off a structure only when you can find it on the model, the cadaver, or the slide without looking at a label. Say the name out loud to your team as you point to it.

Studying the muscles

For every muscle on your list, know three things: where it starts (origin), where it ends (insertion), and what it does (action). The nerve column is there for reference. Cover the chart and quiz yourself one column at a time.

SCAN TO DRILL THE MUSCLES

Interactive Muscle Charts: I, O, A

Origins, insertions, actions, and innervation, drilled interactively. Use it on your phone between lab days.

medmasterscollaborative.com/muscle-charts-i-o-a-inn

PART A

Lab structure list.

These are the lab structures and concepts to learn for the lab portion of the Module 3 exam. Be prepared to identify structures on models, images, the cadaver, and slides.

CHAPTER 1

Muscle Tissue and Microanatomy

SLIDES AND THE SARCOMERE MODEL

Three tissues on the slide, then the sarcomere on the model. Fill in function and location for each tissue as you go; those columns are what the practical asks you to reason from.

☐ Skeletal muscle, longitudinal section

Function _____ **Location** _____

Special structures _____

STRUCTURES TO IDENTIFY

- ☐ Muscle fiber
- ☐ Sarcolemma
- ☐ Sarcoplasm
- ☐ Peripheral nuclei
- ☐ Striations
- ☐ Myofibrils
- ☐ Endomysium

☐ Skeletal muscle, cross section

STRUCTURES TO IDENTIFY

- ☐ Muscle fiber, cut across
- ☐ Fascicle
- ☐ Endomysium
- ☐ Perimysium
- ☐ Epimysium
- ☐ Peripheral nuclei

☐ Cardiac muscle

Function _____ **Location** _____

Special structures _____

STRUCTURES TO IDENTIFY

- ☐ Cardiomyocyte
- ☐ Branching fibers
- ☐ Central nucleus
- ☐ Intercalated discs
- ☐ Striations

☐ Smooth muscle

Function _____ **Location** _____

Special structures _____

STRUCTURES TO IDENTIFY

- ☐ Spindle-shaped cell
- ☐ Central nucleus
- ☐ Absence of striations
- ☐ Circular layer

☐ The muscle organ model

STRUCTURES TO IDENTIFY

- ☐ Epimysium
- ☐ Perimysium
- ☐ Endomysium
- ☐ Fascicle
- ☐ Muscle fiber
- ☐ Tendon
- ☐ Belly
- ☐ Origin

☐ The muscle fiber model

STRUCTURES TO IDENTIFY

- ☐ Sarcolemma
- ☐ Sarcoplasm
- ☐ Myonuclei
- ☐ Myofibril
- ☐ Transverse (T) tubule
- ☐ Sarcoplasmic reticulum
- ☐ Terminal cisternae
- ☐ Triad
- ☐ Mitochondria

☐ The sarcomere model

Be prepared to state which filaments are present in each band, line, and zone.

STRUCTURES TO IDENTIFY

- ☐ Z disc
- ☐ A band
- ☐ I band
- ☐ H zone
- ☐ M line
- ☐ Thick filament (myosin)
- ☐ Thin filament (actin)

Which regions of the sarcomere narrow when a muscle shortens?

Which region never changes length, and why?

CHAPTER 2

Muscles of the Head and Neck

FACE AND NECK MUSCULATURE

The face muscles first, then the two neck muscles. The Head and Neck Muscles lab sprint drills the same list.

☐ Facial expression and mastication

Be prepared to say which of these muscles insert into the modiolus, the corner of the mouth.

STRUCTURES TO IDENTIFY

- ☐ Depressor anguli oris
- ☐ Levator anguli oris
- ☐ Depressor labii inferioris
- ☐ Levator labii superioris
- ☐ Orbicularis oris
- ☐ Orbicularis oculi
- ☐ Risorius
- ☐ Zygomaticus major
- ☐ Zygomaticus minor
- ☐ Nasalis
- ☐ Procerus
- ☐ Temporalis
- ☐ Buccinator
- ☐ Masseter
- ☐ Mentalis
- ☐ Frontal belly of the occipitofrontalis
- ☐ Occipital belly of the occipitofrontalis
- ☐ Galea aponeurotica (epicranial aponeurosis)

☐ Neck

For each muscle, know its origin, insertion and action.

STRUCTURES TO IDENTIFY

- ☐ Platysma
- ☐ Sternocleidomastoid

CHAPTER 3

Muscles of the Back and Thorax

TRUNK MUSCULATURE

Work these in layers, superficial to deep. For every muscle be prepared to give the origin, the insertion, and the action, not just the name.

☐ Superficial back

For each muscle, know its origin, insertion and action.

STRUCTURES TO IDENTIFY

- ☐ Trapezius
- ☐ Latissimus dorsi
- ☐ Levator scapulae
- ☐ Rhomboid major
- ☐ Rhomboid minor
- ☐ Thoracolumbar fascia
- ☐ Ligamentum nuchae

☐ Thoracic wall

Be prepared to state the fiber direction of the external and internal intercostals and what each does to the ribs.

STRUCTURES TO IDENTIFY

- ☐ External intercostals
- ☐ Internal intercostals
- ☐ Intercostal space
- ☐ Neurovascular bundle in the costal groove

☐ The diaphragm

Name what passes through each of the three openings.

STRUCTURES TO IDENTIFY

- ☐ Central tendon
- ☐ Caval opening (T8)
- ☐ Esophageal hiatus (T10)
- ☐ Aortic hiatus (T12)
- ☐ Phrenic nerve

☐ Anterior thorax

STRUCTURES TO IDENTIFY

- ☐ Pectoralis major
- ☐ Pectoralis minor
- ☐ Serratus anterior

CHAPTER 4

Muscles of the Upper Extremity

SHOULDER, ARM, FOREARM, HAND

Learn these by compartment. On a cadaver or a model the compartment tells you the action and the nerve before you have identified the individual muscle.

☐ Rotator cuff

Name the tubercle each one reaches and the movement each one produces.

STRUCTURES TO IDENTIFY

- ☐ Supraspinatus
- ☐ Infraspinatus
- ☐ Teres minor
- ☐ Subscapularis

☐ Scapulohumeral, other

STRUCTURES TO IDENTIFY

- ☐ Deltoid, anterior fibers
- ☐ Deltoid, middle fibers
- ☐ Deltoid, posterior fibers
- ☐ Teres major
- ☐ Coracobrachialis

☐ Arm, anterior compartment

STRUCTURES TO IDENTIFY

- ☐ Biceps brachii, long head
- ☐ Biceps brachii, short head
- ☐ Brachialis
- ☐ Coracobrachialis

☐ Arm, posterior compartment

STRUCTURES TO IDENTIFY

- ☐ Triceps brachii, long head
- ☐ Triceps brachii, lateral head
- ☐ Triceps brachii, medial head
- ☐ Anconeus

☐ Forearm, anterior compartment

STRUCTURES TO IDENTIFY

- ☐ Brachioradialis
- ☐ Pronator teres
- ☐ Flexor carpi radialis
- ☐ Palmaris longus
- ☐ Flexor carpi ulnaris
- ☐ Flexor digitorum superficialis
- ☐ Flexor digitorum profundus
- ☐ Flexor pollicis longus
- ☐ Pronator quadratus
- ☐ Medial epicondyle, common flexor origin

☐ Forearm, posterior compartment

STRUCTURES TO IDENTIFY

- ☐ Extensor carpi radialis longus
- ☐ Extensor carpi radialis brevis
- ☐ Extensor digitorum
- ☐ Extensor digiti minimi
- ☐ Extensor carpi ulnaris
- ☐ Supinator
- ☐ Abductor pollicis longus
- ☐ Extensor pollicis brevis
- ☐ Extensor pollicis longus
- ☐ Lateral epicondyle, common extensor origin

☐ Hand, intrinsic muscles

STRUCTURES TO IDENTIFY

- ☐ Abductor pollicis brevis

☐ Landmarks and retinacula

STRUCTURES TO IDENTIFY

- ☐ Cubital fossa
- ☐ Carpal tunnel
- ☐ Flexor retinaculum
- ☐ Extensor retinaculum

Name the two common origins that most superficial forearm muscles share.

CHAPTER 5

The Heart

MODEL, CADAVER, AND SLIDE

Work the heart from the outside in: sac, wall, chambers, valves, then the vessels on its surface. Be prepared to trace blood through it in order on any specimen you are given.

☐ Position and coverings

STRUCTURES TO IDENTIFY

- ☐ Mediastinum
- ☐ Apex
- ☐ Base
- ☐ Fibrous pericardium
- ☐ Parietal serous pericardium
- ☐ Visceral serous pericardium (epicardium)
- ☐ Pericardial cavity

☐ The heart wall

Function _____ **Location** _____

Special structures _____

STRUCTURES TO IDENTIFY

- ☐ Epicardium
- ☐ Myocardium
- ☐ Endocardium

☐ External surface features

STRUCTURES TO IDENTIFY

- ☐ Anterior interventricular sulcus
- ☐ Posterior interventricular sulcus
- ☐ Right auricle
- ☐ Left auricle
- ☐ Ligamentum arteriosum

☐ Chambers and internal features

STRUCTURES TO IDENTIFY

- ☐ Right atrium
- ☐ Right ventricle
- ☐ Left atrium
- ☐ Left ventricle
- ☐ Interatrial septum
- ☐ Interventricular septum
- ☐ Fossa ovalis
- ☐ Pectinate muscles
- ☐ Trabeculae carneae
- ☐ Papillary muscles
- ☐ Chordae tendineae

☐ Valves

For each valve, name what it separates and what it prevents.

STRUCTURES TO IDENTIFY

- ☐ Tricuspid valve
- ☐ Mitral (bicuspid) valve
- ☐ Pulmonary semilunar valve
- ☐ Aortic semilunar valve

☐ Great vessels

STRUCTURES TO IDENTIFY

- ☐ Superior vena cava
- ☐ Inferior vena cava
- ☐ Pulmonary trunk
- ☐ Right pulmonary artery
- ☐ Left pulmonary artery
- ☐ Pulmonary veins
- ☐ Ascending aorta
- ☐ Aortic arch
- ☐ Descending thoracic aorta

☐ Coronary circulation

STRUCTURES TO IDENTIFY

- ☐ Right coronary artery
- ☐ Left coronary artery
- ☐ Anterior interventricular artery (LAD)
- ☐ Circumflex artery
- ☐ Coronary sinus
- ☐ Great cardiac vein

☐ Conduction system, on the model

STRUCTURES TO IDENTIFY

- ☐ SA node
- ☐ AV node
- ☐ AV bundle (bundle of His)
- ☐ Right bundle branch
- ☐ Left bundle branch
- ☐ Purkinje fibers

☐ **Cardiac muscle slide**

Function _____ **Location** _____

Special structures _____

STRUCTURES TO IDENTIFY

- ☐ Cardiomyocyte
- ☐ Central nucleus
- ☐ Branching fibers
- ☐ Intercalated discs
- ☐ Striations

Trace a drop of blood from the superior vena cava to the aorta, naming every chamber and valve.

_____

_____

CHAPTER 6

Blood Vessels, Histology

SLIDES AND WALL MODELS

Almost every question here is answered from the wall. Identify the tunics first, then use their proportions to name the vessel.

☐ **Artery, cross section**

Function _____ **Location** _____

Special structures _____

STRUCTURES TO IDENTIFY

- ☐ Tunica intima
- ☐ Endothelium
- ☐ Basement membrane
- ☐ Internal elastic lamina
- ☐ Tunica media
- ☐ Smooth muscle
- ☐ External elastic lamina
- ☐ Tunica externa
- ☐ Vasa vasorum
- ☐ Lumen

☐ **Vein, cross section**

Function _____ **Location** _____

Special structures _____

STRUCTURES TO IDENTIFY

- ☐ Tunica intima
- ☐ Tunica media
- ☐ Tunica externa
- ☐ Venous valve
- ☐ Lumen

Name the two features you would use first to tell an artery from a vein on a slide.

_____

☐ **Elastic artery**

Name two elastic arteries you would expect to see labelled.

STRUCTURES TO IDENTIFY

- ☐ Elastic lamellae
- ☐ Thick tunica media
- ☐ Internal elastic lamina
- ☐ External elastic lamina

☐ **Arteriole**

STRUCTURES TO IDENTIFY

- ☐ Arteriole

☐ Fetal circulation, on the model

Name the adult remnant of each fetal structure.

STRUCTURES TO IDENTIFY

- ☐ Foramen ovale
- ☐ Ductus arteriosus

CHAPTER 7

Arteries and Veins of the Thorax and Upper Limb

NAMED VESSELS, CADAVER AND MODEL

These are routes. Trace each one continuously with your finger before you try to name segments, and learn the landmark where each name changes.

☐ Aortic arch and its branches

Be prepared to explain why the right and left sides are not symmetrical.

STRUCTURES TO IDENTIFY

- ☐ Ascending aorta
- ☐ Aortic arch
- ☐ Brachiocephalic trunk
- ☐ Right common carotid artery
- ☐ Right subclavian artery
- ☐ Left common carotid artery
- ☐ Left subclavian artery
- ☐ Descending thoracic aorta

☐ Arteries of the upper limb

Name the landmark at which the subclavian becomes the axillary, and the axillary becomes the brachial.

STRUCTURES TO IDENTIFY

- ☐ Subclavian artery
- ☐ Axillary artery
- ☐ Brachial artery
- ☐ Radial artery
- ☐ Ulnar artery

☐ Superficial veins of the upper limb

STRUCTURES TO IDENTIFY

- ☐ Cephalic vein
- ☐ Basilic vein
- ☐ Median cubital vein

☐ Deep veins of the thorax and upper limb

STRUCTURES TO IDENTIFY

- ☐ Brachial veins
- ☐ Axillary vein
- ☐ Subclavian vein
- ☐ Internal jugular vein
- ☐ Brachiocephalic veins
- ☐ Superior vena cava

Why is there one brachiocephalic artery but two brachiocephalic veins?

CHAPTER 8

Nerves of the Upper Limb

THE BRACHIAL PLEXUS

Learn the five terminal branches first, then the places where each one is most easily injured.

☐ Terminal branches

For each nerve, name the compartment it supplies and the deficit its injury produces.

STRUCTURES TO IDENTIFY

- ☐ Musculocutaneous nerve
- ☐ Median nerve
- ☐ Ulnar nerve
- ☐ Radial nerve
- ☐ Axillary nerve

☐ Nerves at their vulnerable points

Name the injury associated with each site and the sign it produces.

STRUCTURES TO IDENTIFY

- ☐ Axillary nerve at the surgical neck of the humerus
- ☐ Radial nerve in the radial groove
- ☐ Ulnar nerve behind the medial epicondyle
- ☐ Median nerve in the carpal tunnel

☐ Thoracic nerves

STRUCTURES TO IDENTIFY

- ☐ Phrenic nerve
- ☐ Sympathetic trunk

CHAPTER 9

Blood

THE SMEAR

One slide, and every question comes off the nucleus. Identify each cell by the shape of its nucleus first and the granules second.

☐ Whole blood, in a spun tube

STRUCTURES TO IDENTIFY

- ☐ Plasma layer
- ☐ Buffy coat
- ☐ Erythrocyte layer
- ☐ Hematocrit

☐ Erythrocytes

Function _____ **Location** _____

Special structures _____

STRUCTURES TO IDENTIFY

- ☐ Biconcave disc
- ☐ Pale central area
- ☐ Absence of a nucleus

☐ Granulocytes

STRUCTURES TO IDENTIFY

- ☐ Neutrophil
- ☐ Multi-lobed nucleus
- ☐ Fine pale granules
- ☐ Eosinophil
- ☐ Bilobed nucleus
- ☐ Coarse red-orange granules
- ☐ Basophil
- ☐ Nucleus obscured by dark granules

☐ Agranulocytes

STRUCTURES TO IDENTIFY

- ☐ Lymphocyte
- ☐ Large round nucleus with thin rim of cytoplasm
- ☐ Monocyte
- ☐ Kidney-shaped nucleus
- ☐ Abundant pale cytoplasm

☐ Platelets and formation

STRUCTURES TO IDENTIFY

- ☐ Platelet (thrombocyte)
- ☐ Megakaryocyte
- ☐ Red bone marrow
- ☐ Reticulocyte

Order the five leukocytes from most to least numerous.

PART B

Muscle charts.

Origin, insertion, action, and nerve for every muscle in Module 3, grouped the way you will find them in lab. The same information is in the interactive muscle charts.

SCAN TO DRILL THESE CHARTS

Interactive Muscle Charts: I, O, A

Quiz yourself on the origin, insertion, action, and nerve for each muscle on your phone.

medmasterscollaborative.com/muscle-charts-i-o-a-inn

Muscles of facial expression

Every muscle of facial expression is supplied by the facial nerve (CN VII). They pull on skin, not on bone, which is why they can make expressions.

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Frontal belly of the occipitofrontalis	Galea aponeurotica (epicranial aponeurosis)	Skin of the eyebrows and forehead	Raises the eyebrows and wrinkles the forehead	Facial nerve (CN VII)
Occipital belly of the occipitofrontalis	Superior nuchal line of the occipital bone and the mastoid region	Galea aponeurotica	Pulls the scalp backward	Facial nerve (CN VII)
Orbicularis oculi	Medial margin of the orbit (frontal bone, maxilla) and the medial palpebral ligament	Skin around the eyelids	Closes the eye, as in blinking and squinting	Facial nerve (CN VII)
Procerus	Fascia over the nasal bone	Skin between the eyebrows	Pulls the medial eyebrows down and wrinkles the bridge of the nose	Facial nerve (CN VII)
Nasalis	Maxilla beside the nose	Nasal cartilages and the bridge of the nose	Compresses the nasal opening and flares the nostrils	Facial nerve (CN VII)
Orbicularis oris	Maxilla, mandible, and the muscles that blend in at the corners of the mouth	Skin and mucosa of the lips	Closes and purses the lips	Facial nerve (CN VII)
Zygomaticus major	Zygomatic bone, lateral surface	Angle (corner) of the mouth	Draws the corner of the mouth up and out, as in smiling	Facial nerve (CN VII)
Zygomaticus minor	Zygomatic bone	Upper lip	Elevates the upper lip	Facial nerve (CN VII)
Levator labii superioris	Maxilla, along the infraorbital margin	Upper lip	Elevates the upper lip	Facial nerve (CN VII)
Levator anguli oris	Maxilla, below the infraorbital foramen	Angle of the mouth	Elevates the corner of the mouth	Facial nerve (CN VII)

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Risorius	Fascia over the masseter	Angle of the mouth	Draws the corner of the mouth sideways	Facial nerve (CN VII)
Buccinator	Alveolar processes of the maxilla and mandible in the molar region	Orbicularis oris at the angle of the mouth	Presses the cheek against the teeth, keeping food between them while chewing	Facial nerve (CN VII)
Depressor anguli oris	Body of the mandible (oblique line)	Angle of the mouth	Pulls the corner of the mouth down, as in frowning	Facial nerve (CN VII)
Depressor labii inferioris	Body of the mandible, below the lower lip	Lower lip	Pulls the lower lip down	Facial nerve (CN VII)
Mentalis	Mandible, below the incisors	Skin of the chin	Raises and wrinkles the chin and pushes out the lower lip	Facial nerve (CN VII)

Muscles of mastication

Temporalis and masseter both close the jaw. Both are supplied by the mandibular division of the trigeminal nerve (CN V3).

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Temporalis	Temporal fossa of the skull	Coronoid process of the mandible	Elevates (closes) and retracts the mandible	Trigeminal nerve, mandibular division (CN V3)
Masseter	Zygomatic arch	Lateral ramus and angle of the mandible	Elevates the mandible; the prime mover for closing the jaw	Trigeminal nerve, mandibular division (CN V3)

Muscles of the neck

Platysma is a thin sheet in the skin of the neck and works like a muscle of facial expression. Sternocleidomastoid crosses the side of the neck from the sternum and clavicle to the mastoid process.

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Platysma	Fascia over the upper pectoralis major and deltoid	Lower border of the mandible and the skin at the corner of the mouth	Tenses the skin of the neck and pulls the lower lip and corner of the mouth down	Facial nerve (CN VII), cervical branch
Sternocleidomastoid	Manubrium of the sternum and the medial clavicle	Mastoid process of the temporal bone	Both sides: flex the neck. One side: tilts the head to the same side and turns the face to the opposite side	Accessory nerve (CN XI)

Superficial muscles of the back

These move the scapula and the arm. All but the trapezius get their main nerve supply from branches of the brachial plexus (levator scapulae also receives C3 and C4).

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Trapezius	Occipital bone, ligamentum nuchae, spinous processes C7 to T12	Lateral clavicle, acromion, spine of the scapula	Upper fibers elevate the scapula, middle fibers retract it, lower fibers depress it; upper and lower fibers together rotate the glenoid upward	Accessory nerve (CN XI)
Latissimus dorsi	Spinous processes T7 to L5, thoracolumbar fascia, iliac crest, lower ribs	Intertubercular sulcus of the humerus	Extends, adducts, and medially rotates the arm	Thoracodorsal nerve

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Levator scapulae	Transverse processes C1 to C4	Superior medial border of the scapula	Elevates the scapula and rotates the glenoid downward	Dorsal scapular nerve and cervical nerves (C3, C4)
Rhomboid major	Spinous processes T2 to T5	Medial border of the scapula, below the spine	Retracts the scapula and rotates the glenoid downward	Dorsal scapular nerve
Rhomboid minor	Spinous processes C7 to T1	Medial border of the scapula, at the spine	Retracts the scapula and rotates the glenoid downward	Dorsal scapular nerve

Muscles of the thoracic wall and the diaphragm

The external and internal intercostals fill the spaces between the ribs, and their fibers run at right angles to each other. The diaphragm is the main muscle of breathing.

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
External intercostals	Inferior border of the rib above	Superior border of the rib below	Elevate the ribs. Fibers run downward and forward, hands in pockets	Intercostal nerves
Internal intercostals	Superior border of the rib below	Inferior border of the rib above	Depress the ribs. Fibers run at right angles to the external layer	Intercostal nerves
Diaphragm	Xiphoid process, inner surface of ribs 7 to 12, and the lumbar vertebrae by two crura	Central tendon	Contracts and flattens, which increases the vertical dimension of the thoracic cavity	Phrenic nerve (C3, C4, C5)

Anterior thoracic muscles

These three act on the pectoral girdle or the arm from the front of the chest.

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Pectoralis major	Medial clavicle, sternum, costal cartilages 1 to 6	Lateral lip of the intertubercular sulcus of the humerus	Flexes, adducts, and medially rotates the arm	Medial and lateral pectoral nerves
Pectoralis minor	Ribs 3 to 5	Coracoid process of the scapula	Depresses and protracts the scapula, and elevates the ribs when the scapula is fixed	Medial pectoral nerve
Serratus anterior	Ribs 1 to 8 or 9, on the lateral chest wall	Anterior surface of the medial border of the scapula	Protracts the scapula and rotates the glenoid upward; holds the scapula against the chest wall	Long thoracic nerve

Rotator cuff and other muscles crossing the shoulder

Supraspinatus, infraspinatus, and teres minor all reach the greater tubercle. Subscapularis is the only cuff muscle on the lesser tubercle.

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Supraspinatus	Supraspinous fossa of the scapula	Greater tubercle of the humerus	Initiates abduction of the arm, the first fifteen degrees	Suprascapular nerve
Infraspinatus	Infraspinous fossa of the scapula	Greater tubercle of the humerus	Laterally rotates the arm	Suprascapular nerve
Teres minor	Lateral border of the scapula	Greater tubercle of the humerus	Laterally rotates the arm	Axillary nerve

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Subscapularis	Subscapular fossa, the anterior surface of the scapula	Lesser tubercle of the humerus	Medially rotates the arm	Upper and lower subscapular nerves
Deltoid	Lateral clavicle, acromion, spine of the scapula	Deltoid tuberosity of the humerus	Anterior fibers flex and medially rotate, middle fibers abduct, posterior fibers extend and laterally rotate	Axillary nerve
Teres major	Inferior angle of the scapula	Medial lip of the intertubercular sulcus	Extends, adducts, and medially rotates the arm	Lower subscapular nerve
Coracobrachialis	Coracoid process of the scapula	Medial surface of the humeral shaft	Flexes and adducts the arm	Musculocutaneous nerve

Muscles of the arm

Anterior compartment: musculocutaneous nerve. Posterior compartment: radial nerve.

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Biceps brachii	Long head from the supraglenoid tubercle, short head from the coracoid process	Radial tuberosity and the bicipital aponeurosis	Flexes the elbow and supinates the forearm; also flexes the arm	Musculocutaneous nerve
Brachialis	Distal anterior humerus	Coronoid process and tuberosity of the ulna	Flexes the elbow, the prime mover regardless of forearm position	Musculocutaneous nerve
Triceps brachii	Long head from the infraglenoid tubercle, lateral and medial heads from the humerus	Olecranon of the ulna	Extends the elbow; the long head also extends the arm	Radial nerve
Anconeus	Lateral epicondyle of the humerus	Lateral olecranon and posterior ulna	Assists extension of the elbow	Radial nerve

Anterior forearm (flexor compartment)

Most start at the medial epicondyle (common flexor origin). Median nerve, except flexor carpi ulnaris and the medial half of flexor digitorum profundus, which are ulnar.

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Pronator teres	Medial epicondyle of the humerus and coronoid process of the ulna	Lateral mid-shaft of the radius	Pronates the forearm and assists elbow flexion	Median nerve
Flexor carpi radialis	Medial epicondyle	Bases of metacarpals 2 and 3	Flexes and abducts the wrist	Median nerve
Palmaris longus	Medial epicondyle	Palmar aponeurosis	Tenses the palmar aponeurosis; absent in many people	Median nerve
Flexor carpi ulnaris	Medial epicondyle and olecranon	Pisiform, hamate, metacarpal 5	Flexes and adducts the wrist	Ulnar nerve
Flexor digitorum superficialis	Medial epicondyle, coronoid process, radius	Middle phalanges of digits 2 to 5	Flexes the proximal interphalangeal joints	Median nerve
Flexor digitorum profundus	Anterior and medial ulna and interosseous membrane	Distal phalanges of digits 2 to 5	Flexes the distal interphalangeal joints	Lateral half: median nerve. Medial half: ulnar nerve
Flexor pollicis longus	Anterior radius and interosseous membrane	Distal phalanx of the thumb	Flexes the thumb	Median nerve (anterior interosseous branch)

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Pronator quadratus	Distal anterior ulna	Distal anterior radius	Pronates the forearm, the prime mover of pronation	Median nerve (anterior interosseous branch)

Posterior forearm (extensor compartment)

Most start at the lateral epicondyle (common extensor origin). Every muscle here is supplied by the radial nerve or its deep branch.

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Brachioradialis	Lateral supracondylar ridge of the humerus	Styloid process of the radius	Flexes the elbow, strongest with the forearm in mid-position	Radial nerve
Extensor carpi radialis longus	Lateral supracondylar ridge	Base of metacarpal 2	Extends and abducts the wrist	Radial nerve
Extensor carpi radialis brevis	Lateral epicondyle	Base of metacarpal 3	Extends and abducts the wrist	Radial nerve, deep branch
Extensor digitorum	Lateral epicondyle	Extensor expansions of digits 2 to 5	Extends the fingers and the wrist	Radial nerve, deep branch (posterior interosseous)
Extensor digiti minimi	Lateral epicondyle	Extensor expansion of digit 5	Extends the little finger	Radial nerve, deep branch (posterior interosseous)
Extensor carpi ulnaris	Lateral epicondyle and posterior ulna	Base of metacarpal 5	Extends and adducts the wrist	Radial nerve, deep branch (posterior interosseous)
Supinator	Lateral epicondyle and proximal ulna	Proximal lateral radius	Supinates the forearm	Radial nerve, deep branch
Abductor pollicis longus	Posterior radius, ulna, interosseous membrane	Base of metacarpal 1	Abducts and extends the thumb	Radial nerve, deep branch (posterior interosseous)
Extensor pollicis brevis	Posterior radius and interosseous membrane	Proximal phalanx of the thumb	Extends the thumb at the metacarpophalangeal joint	Radial nerve, deep branch (posterior interosseous)
Extensor pollicis longus	Posterior ulna and interosseous membrane	Distal phalanx of the thumb	Extends the thumb at the interphalangeal joint	Radial nerve, deep branch (posterior interosseous)

Intrinsic muscle of the hand

Abductor pollicis brevis is the only hand muscle on your structure list. Find it in the fleshy pad at the base of the thumb.

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Abductor pollicis brevis	Flexor retinaculum, scaphoid, and trapezium	Base of the proximal phalanx of the thumb	Abducts the thumb	Median nerve (recurrent branch)

PART C

Blood cell charts.

Identify every cell by the shape of its nucleus first and its granules second. The counts are typical adult ranges; labs vary slightly.

The formed elements, part 1: what it is

CELL	WHAT IT LOOKS LIKE	CELL LINE	GRANULOCYTE OR AGRANULOCYTE	HOW MANY	MAIN JOB
Erythrocyte (red blood cell)	Biconcave disc about 7.5 micrometers across with a pale center. No nucleus and no organelles; packed with hemoglobin.	Myeloid	Neither (not a leukocyte)	About 4 to 6 million per microliter; the most numerous formed element	Carries oxygen to the tissues and some carbon dioxide back, bound to hemoglobin
Neutrophil	Nucleus with three to five lobes joined by thin strands. Fine, pale lilac granules. About twice the size of a red cell.	Myeloid	Granulocyte	50 to 70 percent of white cells; the most numerous leukocyte	First responder. Leaves the blood and engulfs bacteria (phagocytosis)
Eosinophil	Nucleus with two lobes. Large red-orange granules that fill the cytoplasm.	Myeloid	Granulocyte	2 to 4 percent of white cells	Attacks parasitic worms and helps control allergic reactions
Basophil	Lobed nucleus that is usually hidden under large, dark blue-purple granules.	Myeloid	Granulocyte	Less than 1 percent of white cells; the least numerous leukocyte	Releases histamine, which widens blood vessels during inflammation and allergy, and heparin, which slows clotting
Lymphocyte	Large, round, dark-staining nucleus that fills most of the cell, with a thin rim of pale blue cytoplasm. Usually a little larger than a red cell.	Lymphoid	Agranulocyte	20 to 40 percent of white cells; the second most numerous leukocyte	Specific immunity. T cells attack infected cells directly; B cells become plasma cells that make antibodies
Monocyte	The largest white cell. Kidney- or horseshoe-shaped nucleus with abundant pale gray-blue cytoplasm.	Myeloid	Agranulocyte	2 to 8 percent of white cells	Leaves the blood and becomes a macrophage that engulfs pathogens and debris
Platelet (thrombocyte)	Small purple cell fragment, 2 to 4 micrometers, with no nucleus. Often seen in small clumps between the red cells.	Myeloid (breaks off a megakaryocyte)	Neither (a cell fragment)	150,000 to 400,000 per microliter	Forms the platelet plug and helps start clotting when a vessel is damaged

The formed elements, part 2: too many and too few

The term for a high count usually ends in -philia or -cytosis. The term for a low count ends in -penia.

CELL	ELEVATED WHEN	TERM	DECREASED WHEN	TERM
Erythrocyte (red blood cell)	Living at high altitude, chronic lung disease, dehydration, or a marrow disorder (polycythemia vera)	**Polycythemia**	Blood loss, iron or vitamin B12 deficiency, kidney disease, destruction of red cells, or marrow failure	**Anemia**
Neutrophil	Acute bacterial infection, inflammation, tissue injury, or stress	**Neutrophilia**	Chemotherapy, some viral infections, some drugs, or marrow suppression	**Neutropenia**
Eosinophil	Parasitic worm infection, allergies, and asthma	**Eosinophilia**	Corticosteroid drugs or high cortisol from stress	**Eosinopenia**
Basophil	Allergic reactions and some leukemias, such as chronic myeloid leukemia	**Basophilia**	Corticosteroid drugs, stress, or an overactive thyroid (hard to measure because the normal count is so low)	**Basopenia**

CELL	ELEVATED WHEN	TERM	DECREASED WHEN	TERM
Lymphocyte	Viral infections such as mononucleosis, and chronic lymphocytic leukemia	**Lymphocytosis**	HIV infection, corticosteroid drugs, chemotherapy, or radiation	**Lymphopenia**
Monocyte	Chronic infections such as tuberculosis, and chronic inflammation	**Monocytosis**	Corticosteroid drugs or marrow suppression	**Monocytopenia**
Platelet (thrombocyte)	Inflammation, iron deficiency, after the spleen is removed, or a marrow disorder	**Thrombocytosis**	Leukemia, chemotherapy, some drugs, viral infection, or an enlarged spleen that traps platelets	**Thrombocytopenia**

Where the cells come from

CELL	WHAT IT LOOKS LIKE	WHAT TO KNOW
Megakaryocyte	Very large marrow cell with a big, many-lobed nucleus	Breaks into thousands of platelets; stays in the red bone marrow
Reticulocyte	Young red cell that has just lost its nucleus; a network of leftover ribosomes shows with a special stain	About 1 percent of red cells. The count rises after blood loss as the marrow speeds up (reticulocytosis) and falls when the marrow fails

Memory line for the white cells from most to least numerous: neutrophil, lymphocyte, monocyte, eosinophil, basophil ("Never Let Monkeys Eat Bananas").

Blood terms

Hematocrit
The percent of blood volume made up of red cells after the blood is spun. Roughly 42 to 52 percent in males and 37 to 47 percent in females.

Plasma
The liquid part of blood, about 55 percent of the volume. Mostly water, plus proteins (albumin, globulins, fibrinogen), nutrients, gases, and wastes.

Hematopoiesis
Formation of all blood cells, in the red bone marrow. Erythropoiesis makes red cells, leukopoiesis makes white cells, and thrombopoiesis makes platelets.

Lymphoid line
The stem cell line that makes lymphocytes.

Agranulocyte
A white cell without visible granules: lymphocyte and monocyte.

Leukopenia
A white cell count below normal (below about 4,500 per microliter).

-philia and -cytosis
Word endings that mean too many of that cell, as in neutrophilia or lymphocytosis.

Buffy coat
The thin whitish layer between the plasma and the red cells in a spun tube. It holds the white cells and platelets, less than 1 percent of the volume.

Formed elements
The cells and cell fragments of blood: erythrocytes, leukocytes, and platelets.

Myeloid line
The stem cell line that makes red cells, platelets, neutrophils, eosinophils, basophils, and monocytes.

Granulocyte
A white cell with visible granules in its cytoplasm: neutrophil, eosinophil, basophil.

Leukocytosis
A white cell count above normal (above about 11,000 per microliter), usually from infection or inflammation.

Diapedesis
A white cell squeezing out of a capillary between the endothelial cells to reach the tissue.

-penia
A word ending that means too few of that cell, as in neutropenia or thrombocytopenia.

Anemia
Too little oxygen-carrying capacity: too few red cells or too little hemoglobin.

Polycythemia
Too many red cells, which thickens the blood.

Hemophilia
An inherited lack of a clotting factor, so blood clots slowly.

Sickle cell disease
An inherited change in hemoglobin that bends red cells into a sickle shape when oxygen is low.

Leukemia
A cancer of white cell-forming tissue that fills the marrow with abnormal white cells.

PART D

Questions you could be asked on the lab exam.

At each station a structure will be pinned or pointed to. These are the kinds of questions that go with it. Practice answering them out loud with a partner, with the structure in front of you.

Muscle tissue and microanatomy

1. Identify the tissue on this slide: skeletal, cardiac, or smooth muscle.
2. Name the structure at the pointer (for example, intercalated disc, peripheral nucleus, or striation).
3. Is this skeletal muscle cut longitudinally or in cross section?
4. Name the connective tissue layer at the pointer: epimysium, perimysium, or endomysium.
5. What does the connective tissue layer at the pointer surround?
6. Name the band, line, or zone of the sarcomere at the pointer.
7. Which filaments are found in the region at the pointer: thick, thin, or both?
8. Name the structure on the muscle fiber model: T tubule, terminal cisterna, sarcoplasmic reticulum, or triad.
9. Which sarcomere regions get narrower when the muscle contracts, and which one stays the same?

Muscles of the head and neck

1. Identify the muscle at the pointer.
2. What expression does the muscle at the pointer produce?
3. Is the muscle at the pointer a muscle of facial expression or a muscle of mastication?
4. Where does the muscle at the pointer insert?
5. Name the muscle that divides the neck into its anterior and posterior triangles.
6. What happens to the head when only the left sternocleidomastoid contracts?

Muscles of the back and thorax

1. Identify the muscle at the pointer.
2. Give the origin, insertion, or action of the muscle at the pointer.
3. Which direction do the fibers of the external intercostals run, and what do they do to the ribs?
4. What structures pass through the caval opening, the esophageal hiatus, and the aortic hiatus, and at which vertebral levels?
5. Name the muscle that attaches to the coracoid process at the pointer.

Muscles of the upper extremity

1. Identify the muscle at the pointer.
2. Give the origin, insertion, or action of the muscle at the pointer.
3. Which rotator cuff muscle attaches to the lesser tubercle?
4. Name the tubercle this rotator cuff muscle reaches and the movement it produces.
5. Is this forearm muscle in the anterior or posterior compartment?
6. Name the common origin shared by the muscles at the pointer: medial or lateral epicondyle.
7. Which muscle flexes the elbow in every forearm position, and which is the prime mover of pronation?
8. Name the landmark at the pointer: cubital fossa, carpal tunnel, flexor retinaculum, or extensor retinaculum.

The heart

1. Identify the chamber, valve, or vessel at the pointer.
2. What two chambers, or which chamber and vessel, does this valve separate?
3. Name the layer of the heart wall or pericardium at the pointer.
4. Identify the internal feature at the pointer: pectinate muscles, trabeculae carneae, papillary muscle, chordae tendineae, or fossa ovalis.
5. Name the coronary artery or cardiac vein at the pointer.
6. Where does the coronary sinus empty?
7. Trace a drop of blood from the superior vena cava to the aorta, naming every chamber and valve.
8. Name the part of the conduction system at the pointer.
9. What fetal structure became the ligamentum arteriosum, and what did it connect?

Blood vessels

1. Is this vessel an artery or a vein? Name one feature you used to decide.
2. Name the tunic or layer at the pointer.
3. Is this an elastic artery or an arteriole?
4. Identify the named artery or vein at the pointer.
5. At which landmark does the subclavian artery become the axillary artery? The axillary become the brachial?
6. Name the three branches of the aortic arch in order from right to left.
7. Name the vein most often used to draw blood and where it lies.
8. Name the adult remnant of the fetal structure at the pointer.

Nerves of the upper limb

1. Name the nerve at the pointer.
2. Name the nerve injured at this site and the sign it produces (for example, wrist drop or claw hand).
3. Which nerve runs behind the medial epicondyle?
4. Which nerve wraps around the surgical neck of the humerus?
5. Which nerve passes through the carpal tunnel?

Blood

1. Identify the cell at the pointer.
2. Is the cell at the pointer a granulocyte or an agranulocyte?
3. Is the cell at the pointer from the myeloid line or the lymphoid line?
4. What is the main job of the cell at the pointer?
5. Name the layer at the pointer in the spun tube: plasma, buffy coat, or red cells.
6. What does the hematocrit measure?
7. Put the five white cells in order from most to least numerous.
8. Which white cell would you expect to be high in a parasitic worm infection? In an acute bacterial infection? In mononucleosis?
9. What is the term for too few platelets? Too many white cells?
10. Name the cell that breaks apart to form platelets.